Advanced Scraping: Dynamic Websites

Fall 2025

bit.ly/25-advanced-scraping

Description

Most modern websites feed data asynchronously to the page based on user interaction. This makes such sites not only dynamic and fast, but also a challenge to scrape. This 5-week module digs into acquiring data from sites that reveal data only after interactions with pulldown menus, scrolling, clicking, searching and other inputs. Once you have completed this module, you will be able to scrape just about any website you target.

Instructor

Sandeep Junnarkar is the director of Data Journalism Journalism (and a founding faculty member) at the Newmark Graduate School of Journalism. He also works at Bloomberg News where he's a global newsroom trainer for data journalism and a special projects editor. He formerly worked at The New York Times as a Web editor and writer. Junnarkar has won numerous awards for his investigative reporting. His most recent project, AI Monitors, received a Knight Foundation grant to train community reporters about algorithmic accountability.

Class Recording

Your attendance in class is mandatory but these recordings are meant to provide a way to review the material.

GitHub Repo

Our class repo.

Assignment

You are required to scrape one dynamic website from which you need data for your capstone or major reporting assignment. If you don't have an appropriate site to scrape, I will assign you a challenging one so you can put your newly acquired skills to the test. Post your pitch in a google doc, which you share with me. In your pitch, let me know how this will be useful for either your capstone or some major project you're working on.

Please file your pitch by November 11th @ 10pm.

I will give you feedback on that pitch, and then you can go ahead and scrape it.

The complete scrape is due by December 10th at 10pm.

Week 1 – Static URLS

We uncover where a website is pulling its data from using the advanced features of the Chrome browser's Inspect Element tool. We will test out our newly acquired skills on a site where the URL never changes.

Week 2 – Pure JSON

The most efficient method for web developers to build a page that retrieves massive amounts of data is through JavaScript that taps data stored in the JSON format. BeautifulSoup can't help here. We just need to know how to find the JSON file, to request it and then to parse it. We'll do exactly that in week 3.

Week 3 – Scrolling sites

Infinite scroll sites are designed for the mobile age. Links are hard to tap with a finger on a small device, but a simple swipe easily scrolls the page down to reveal more data. That can make scraping an infinite scroll page difficult. We'll learn to find the actual location of the data buried in the scrolls.

Week 4 –Tapping the Data Pipeline 1

Many modern sites offer an API or Application Programming Interface for people to access their services or data. These sites would rather you access their information "hose" built specifically to tap all their data rather than tax their websites. The challenge is that all APIs are slightly different and you'll have to tackle each one individually. We'll explore one that can serve as your template.

Week 5 – Tapping the Data Pipeline 2

We'll tap APIs that use search, and also work with rate limits/paginations.